or scheduling constraints. Mitigate this through bounded formulations, coefficient-sensitivity analysis, and independent classical feasibility checks against the original model—not only the QUBO objective.

Hardware and statistical uncertainty: Noise, limited connectivity, calibration drift, and finite sampling can produce variable results. Characterise performance across repeated runs, report uncertainty, and retain device and compilation metadata. Error mitigation must not be confused with fault tolerance. Below-threshold quantum-memory demonstrations are important progress, but do not establish reliable robotics optimisation or application-level advantage.

Deadline and availability risk: Remote execution may miss planning deadlines because of queueing, communication failure, or repeated optimization. Timestamp requests, reject stale results, and bound retries and execution budgets. Maintain a continuously available classical planner; quantum-service failure must not interrupt feedback control, emergency stopping, or protective monitoring.

***Model and physical-safety risk*:** A feasible mathematical plan can fail under uncertain payloads, friction, localisation, or moving obstacles. Validate candidates in simulation with disturbance scenarios, then use staged hardware testing and runtime monitoring. Simulation is a screening mechanism, not proof of safety. Apply conservative margins and revalidate before execution when conditions change.

Security and evidence risk: Cloud submissions may expose layouts, production priorities, or operational constraints. Minimise submitted information, enforce access controls, encrypt transfers, and audit solver interfaces. Benchmark against strong classical methods, including preprocessing and validation costs. Report negative results and separate classical improvements from quantum contributions. Avoid unsupported claims concerning quantum-trained robot policies or quantum sensing as evidence of computational advantage. END

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